

Gradient Descent for Linear Regression

Linear Regression Model

$$f_{w,b}(x) = wx + b$$

Cost function

$$J(w,b) = \frac{1}{2m} \sum_{i=1}^n (f_{w,b}(x^{(i)}) - y^{(i)})^2$$

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Gradient Descent Algo:

until convergence {

$$W = W - \alpha \frac{\partial J(w,b)}{\partial w} \rightarrow \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)}) x^{(i)} \quad (i)$$
$$b = b - \alpha \frac{\partial J(w,b)}{\partial b} \rightarrow \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)}) \quad (ii)$$

}

Derivation of both (i) & (ii):

$$J(w,b) = \frac{1}{2m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2$$

Step 1: Substitute $f(x)$:

$$J(w,b) = \frac{1}{2m} \sum (wx^{(i)} + b - y^{(i)})^2$$

Step 2: apply chain rule:

chain rule says: if you have (something)², its derivative is 2 * (something) * (derivative of that something)

→ let $\text{error}^{(i)} = wx^{(i)} + b - y^{(i)}$

So we are differentiating $(\text{error}^{(i)})^2$ with r.t. w :

$$\frac{d}{dw} (\text{error}^{(i)})^2 = 2 \cdot \text{error}^{(i)} \cdot \frac{d}{dw} (\text{error}^{(i)})$$

Step 3 $\rightarrow$ find $\frac{d}{dw} (\text{error}^{(i)})$.

Take derivative w.r. to w , treating $b, x^{(i)}, y^{(i)}$ as constants (since we are only asking "how this change if I wiggle w):

- derivative of $w x^{(i)}$ w.r. to $w \rightarrow x^{(i)}$
- " " b w.r. to $w \rightarrow 0$
- " " $-y^{(i)}$ " " " $w \rightarrow 0$

Step 4: put together:

$$\frac{d}{dw} (\text{error}^{(i)})^2 = 2 \cdot \text{error}^{(i)} \cdot x^{(i)}$$

Step 5: plug back:

$$\frac{dJ}{dw} = \frac{1}{2m} \sum [2 \cdot \text{error}^{(i)} \cdot x^{(i)}]$$

$$\frac{dJ}{dw} = \frac{1}{m} \sum (f(x^{(i)}) - y^{(i)}) \cdot x^{(i)}$$

Same steps for b :

as in S3 for this you will treat $w, x^{(i)}, y^{(i)}$ as constants, and will get

derivative of $b = 1$; both 0 so, no $x^{(i)}$

(Optional)

$$\frac{\partial}{\partial w} J(w, b) = \frac{\partial}{\partial w} \frac{1}{2m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2 = \frac{\partial}{\partial w} \frac{1}{2m} \sum_{i=1}^m (\underline{wx^{(i)} + b} - y^{(i)})^2$$

$$= \cancel{\frac{1}{2m}} \sum_{i=1}^m (\underline{wx^{(i)} + b} - y^{(i)}) \cancel{2} x^{(i)} = \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)}) x^{(i)}$$

$$\frac{\partial}{\partial b} J(w, b) = \frac{\partial}{\partial b} \frac{1}{2m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2 = \frac{\partial}{\partial b} \frac{1}{2m} \sum_{i=1}^m (\underline{wx^{(i)} + b} - y^{(i)})^2$$

$$= \cancel{\frac{1}{2m}} \sum_{i=1}^m (\underline{wx^{(i)} + b} - y^{(i)}) \cancel{2} = \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})$$

no $x^{(i)}$

Now you have these two

So Gradient Descent algorithm be:

$$\begin{aligned} & \text{repeat until convergence } \left\{ \begin{array}{l} w = w - \alpha \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)}) x^{(i)} \rightarrow \frac{dJ(w,b)}{dw} \\ b = b - \alpha \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)}) \rightarrow \frac{dJ(w,b)}{db} \end{array} \right. \\ & \} \end{aligned}$$

We saw,

Gradient descent lead to local minimum instead of global minimum.

→ point with lowest possible value of all the points

In sq. error cost func.

the bowl one, there is only one global minimum here.

→ this c.f is a convex function
